

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                                                                                             | (Level/CO)           | Marks     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------|
| <b>Q. 1 Solve <u>any two</u> of the following.</b>                                                                                                                                                                                                                                                                                                                                                                          |                      | <b>12</b> |
| A) A hollow cylinder 2 m long has an outside diameter of 50 mm and inside diameter of 30 mm. If the cylinder is carrying a load of 25 kN, find the stress in the cylinder. Also find the deformation of the cylinder, if the value elastic modulus for the cylinder material is 100 GPa.                                                                                                                                    | Understand/<br>CO3   | <b>06</b> |
| B) A steel bar 2 m long and 40 mm in diameter is subjected to an axial pull of 80 kN. Find the length of the 20 mm diameter bore, which should be centrally carried out, so that the total elongation should increase by 20 % under the same pull. Take E for the bar material as 200 GPa.                                                                                                                                  | Apply/ CO3           | <b>06</b> |
| C) Two parallel walls 6 m apart are stayed together by a steel rod 25 mm diameter passing through metal plates and nuts at each end. The nuts are tightened, when the rod is at a temperature of 100° C. Determine the stress in the rod, when the temperature falls down to 60° C, if<br>a) The ends do not yield, and<br>b) The ends yield by 1 mm<br>Take E = 200 GPa and $\alpha = 12 \times 10^{-6} / ^\circ \text{C}$ | Analyze/<br>CO3      | <b>06</b> |
| <b>Q.2 Solve <u>any two</u> following questions.</b>                                                                                                                                                                                                                                                                                                                                                                        |                      | <b>12</b> |
| A) Define and explain the below terms:<br>A.1) Shear Force (SF),<br>A.2) Bending Moment (BM)<br>A.3) Relation between loading, SF, and BM                                                                                                                                                                                                                                                                                   | Understand<br>/ CO 2 | <b>06</b> |
| B) Draw the <b>shear force and bending moment</b> diagrams for a cantilever beam of span 1.5 m carrying point loads of 2 kN and 1.5 kN at distance of 1 m and 1.5 from fixed end respectively.                                                                                                                                                                                                                              | Apply /<br>CO 2      | <b>06</b> |
| C) A simply supported beam 6 m long is carrying a uniformly distributed load of 5 kN/m over a length of 3 m from the right end. Draw the <b>SF and BM diagrams</b> for the beam and also calculate the <b>maximum BM</b> on the section.                                                                                                                                                                                    | Analyze /<br>CO 2    | <b>06</b> |

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| <b>Q. 3</b> | <b>Solve <u>any two</u> of the following.</b>                                                                                                                                                                                                                                                                                      |                    | <b>12</b> |
| A)          | State and explain the assumptions in the theory of simple bending.                                                                                                                                                                                                                                                                 | Understand/<br>CO2 | 06        |
| B)          | A cantilever beam is rectangular in section having 80 mm width and 120 mm depth. If the cantilever is subjected to a point load of 6 kN at the free end and the bending stress is not to exceed 40 MPa, find the span of the cantilever beam.                                                                                      | Apply / CO2        | 06        |
| C)          | Calculate the maximum torque that a shaft of 125 mm diameter can transmit, if the maximum angle of twist is 10 in a length of 1.5 m. Take $C = 70$ GPa.                                                                                                                                                                            | Analyze /<br>CO2   | 06        |
| <b>Q.4</b>  | <b>Solve <u>any two</u> of the following.</b>                                                                                                                                                                                                                                                                                      |                    | <b>12</b> |
| A)          | State and explain the assumption in the Euler's column theory.                                                                                                                                                                                                                                                                     | Remember/<br>CO3   | 06        |
| B)          | An I section joist 400 mm x 200 mm x 20 mm and 6 m long is used as a strut with both ends fixed. What is Euler's crippling load for the column? Take elastic modulus for the joist as 200 GPa.                                                                                                                                     | Apply /<br>CO3     | 06        |
| C)          | Find the <b>Euler's crippling load</b> for a hollow cylindrical steel column of 38 mm external diameter and 2.5 mm thick. Take length of the column as 2.3 and hinged at its both ends. Take $E = 205$ GPa.<br>Also, <b>determine crippling load</b> by Rankine's formula using constants as $1/7500$ and yield stress as 335 MPa. | Apply / CO3        | 06        |
| <b>Q. 5</b> | <b>Solve <u>any two</u> of the following.</b>                                                                                                                                                                                                                                                                                      |                    | <b>12</b> |
| A)          | Define and explain in detail below terms:<br>A.1) Principal planes<br>A.2) Principal stresses                                                                                                                                                                                                                                      | Understand         | 06        |
| B)          | A wooden bar is subjected to a tensile stress of 5 MPa. What will be the values of normal and shear stresses across a section, which makes an angle of $25^\circ$ with the direction of the tensile stress.                                                                                                                        | Apply              | 06        |
| C)          | Enlist the different theories of failure and explain in details one of the theory.                                                                                                                                                                                                                                                 | Understand         | 06        |

\*\*\* End \*\*\*